Simplex Method.

Q. Solve by simplex method:

Max. $Z = 3x_1 + 2x_2$

Subject to $x_1 + x_2 \leq 4$

$x_1 - x_2 \leq 2$

$x_1, x_2 \geq 0$

Solution:

Converting linear inequalities to linear equality by introducing slack variables.

$\therefore$ The LPP is,

max. $z = 3x_1 + 2x_2 + 0S_1 + 0S_2$

s.t. $x_1 + x_2 + S_1 = 4$

$x_1 - x_2 + S_2 = 2$

$x_1, x_2 \geq 0$, $S_1, S_2 \geq 0$.

Simplex table:

Basic Variable	$c_j \rightarrow$ $\downarrow$	3	2	0	0		
		x_1	x_2	S_1	S_2	b	min Ratio
S_1	0	1	1	1	0	4	$4/1=4$
S_2	0	1	-1	0	1	2	$2/1=2$
	z_j	0	0	0	0		
	z_j-c_j	-3	-2	0	0		
		$\therefore$ x_1 enters & S_2 departs					
S_1	0	0	2	1	-1	2	$2/2=1$
x_1	3	1	-1	0	1	2	—
	z_j	3	-3	0	3		
	z_j-c_j	0	-5	0	3		
		$\therefore$ x_2 enters & S_1 departs					
x_2	2	0	1	$1/2$	$-1/2$	1	
x_1	3	1	0	$1/2$	$1/2$	3.	
	z_j	3	2	$5/2$	$1/2$		
	z_j-c_j	0	0	$5/2$	$1/2$		

As all entries in $(z_j - c_j)$ row ≥ 0.

$\therefore$ optimal solution is reached. &

optimal solution is,

$x_1 = 3$, $x_2 = 1$.

Also

$$Z_{max} = 3 \times 3 + 2 \times 1$$

$$= 9 + 2$$

$$\boxed{Z_{max} = 11 \text{ units}}$$

Q. Solve by Big-M method.

$$\max.\ z = 3x_1 - x_2$$

Subject to $2x_1 + x_2 \geq 2$

$$x_1 + 3x_2 \leq 2$$

$$x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Solution: converting linear inequalities to linear equalities by introducing slack, surplus & artificial variables.

$\therefore$ LPP is,

$$\max.\ z = 3x_1 - x_2 + 0S_1 + 0S_2 + 0S_3 - MA_1$$

s.t. $2x_1 + x_2 - S_1 + A_1 = 2$

$$x_1 + 3x_2 + S_2 = 2$$

$$x_2 + S_3 = 4$$

$$x_1, x_2 \geq 0,\ S_1, S_2, S_3 \geq 0,\ A_1 \geq 0$$

Table :

Basic variable	$C_j \rightarrow$ ↓	3	-1	0	0	0	-m	b	min Ratio
		x_1	x_2	S_1	S_2	S_3	A_1		
A_1	-m	2	1	-1	0	0	1	2	$2/2=1$
S_2	0	1	3	0	1	0	0	2	$2/1=2$
S_3	0	0	1	0	0	1	0	4	—
	z_j	-2m	-m	m	0	0	-m		
	z_j-c_j	(-2m-3)	(-m+1)	m	0	0	0		
		∴ x_1 enters & A_1 departs.							
x_1	3	1	1/2	-1/2	0	0	1/2	1	-
S_2	0	0	5/2	1/2	1	0	-1/2	1	2
S_3	0	0	1	0	0	1	0	4	-
	z_j	3	3/2	-3/2	0	0	3/2		
	z_j-c_j	0	5/2	-3/2	0	0	3/2+m		
		S_1 enters & S_2 departs							
x_1	3	1	~~5/2~~ 3	0	1	0	0	2	
S_1	0	0	5	1	2	0	-1	2	
S_3	0	0	1	0	0	1	0	4	
	z_j	3	9	0	3	0	0		
	z_j-c_j	0	10	0	3	0	0		

∵ All $z_j-c_j \geq 0$

Optimal solution is reached.

∴ $x_1 = 2$ & $x_2 = 0$.

$Z_{max} = 3(2) - 0$

$\boxed{Z_{max} = 6 \text{ units}}$